Introduction to Numerical Analysis II, 5044, Spring 2022
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Homework set 2 due 02/21/2025 Name: Logan Reed

This worksheet is adapted from the homework sheets by Dr. Seibold.

Use the worksheet as the cover page and add your name.
Upload a single pdf for all written parts and separate .m-files (1 file per problem).
Please name your .m-files using your last name, the worksheet number, and problem number.

1. (*100 points*) Consider the Lotka-Volterra predator-prey model from the previous worksheet:

$$\frac{d}{dt}\vec{u}(t) = \vec{f}(\vec{u}(t)), \tag{1}$$

where $\vec{u} = \begin{bmatrix} u \\ v \end{bmatrix}$ and $\vec{f}(\vec{u}) = \begin{bmatrix} f_1(u,v) \\ f_2(u,v) \end{bmatrix} = \begin{bmatrix} u - 4uv \\ -v + 5uv \end{bmatrix}$.

(a) Calculate the Jacobian matrix $D\vec{f}(\vec{u}) = \begin{bmatrix} \frac{\partial f_1}{\partial u}(u,v) & \frac{\partial f_1}{\partial v}(u,v) \\ \frac{\partial f_2}{\partial u}(u,v) & \frac{\partial f_2}{\partial v}(u,v) \end{bmatrix}$ and the product $D\vec{f}(\vec{u}) \cdot \vec{f}(\vec{u})$.

(b) Use the expression from (a) to formulate a second order Taylor series for (1).

(c) Start with the results produced in parts (d) and (e) of homework sheet 1, and add the analogous plots produced with the Taylor series method formulated above (using the same parameters, but using different colors, line styles, and/or labels for the two methods). Explain your observations.

(d) Use the Taylor series method to approximate the time T that it takes for a trajectory $\vec{u}(t)$ to return (for the first time) to its initial value, i.e. $\vec{u}(T) = \vec{u}(0)$. Do this for at least 50 initial values

$$\vec{u}(0) = \begin{bmatrix} a \\ 0.25 \end{bmatrix} \text{ with } 0 < a < 0.2$$

and thus produce a plot of the function $T(a)$.

(e) Explain the fact that for $a \approx 0.25$ it takes close to $T = 2\pi$ for the solution to "go around" once.

1a) $f = \begin{bmatrix} g \\ h \end{bmatrix} = \begin{bmatrix} u - 4uv \\ -v + 5uv \end{bmatrix}$, $\quad g_u = 1 - 4v \quad h_u = 5v$

$g_v = -4u \quad h_v = -1 + 5u$

$$Df = \begin{bmatrix} 1-4u & -4u \\ 5v & -1+5u \end{bmatrix}$$

$(1-4u)(u-4uv) = u - 4uv - 4uv + 16uv^2$

$-4u(-v+5uv) = 4uv - 20u^2v$

$5v(u-4uv) = 5uv - 20uv^2$

$(-1+5u)(-v+5uv) = v - 5uv - 5uv + 25u^2v$

$$(Df)f = \begin{bmatrix} 1-4v & -4u \\ 5v & -1+5u \end{bmatrix} \begin{bmatrix} u-4uv \\ -v+5uv \end{bmatrix} = \begin{bmatrix} u - 4uv - 20u^2v + 16uv^2 \\ v - 5uv + 25u^2v - 20uv^2 \end{bmatrix}$$

1b) $\vec{u}(t+k) \approx \vec{u} + k f(\vec{u}) + k^2 (Df) f$

$$= \begin{bmatrix} u \\ v \end{bmatrix} + k \begin{bmatrix} u-4uv \\ -v+5uv \end{bmatrix} + k^2 \begin{bmatrix} u - 4uv - 20u^2v + 16uv^2 \\ v - 5uv + 25u^2v - 20uv^2 \end{bmatrix}$$